

MODULE *records*

Record constructors for the version 2 *Zcash P2P* protocol (draft *ZIP*, zcash/zips#1344). Records are the units carried on the handshake stream and on announcement streams; request/response formats follow in later phases.

Fields that are irrelevant to the properties being verified (nonce, services, user agent, relay preferences) are set to constant values to keep the *TLC* state space manageable, following the same convention as the legacy *messages.tla*.

EXTENDS *Naturals*

The *init* record exchanged on the handshake stream (draft: “Init Record”).

version is the sender’s advertised protocol version and *start_height* its best block height; both are the only fields the model acts on.

$\text{MakeInit}(\text{version}, \text{height}) \triangleq [$
 $\quad \text{kind} \quad \mapsto \text{“init”},$
 $\quad \text{version} \quad \mapsto \text{version},$
 $\quad \text{services} \quad \mapsto 0,$
 $\quad \text{nonce} \quad \mapsto 0,$
 $\quad \text{user_agent} \mapsto \text{“”},$
 $\quad \text{start_height} \mapsto \text{height},$
 $\quad \text{relay} \quad \mapsto 1,$
 $\quad \text{announce} \quad \mapsto 0,$
 $\quad \text{full_ids} \quad \mapsto 0$
 $]$

A header announcement record on the block announcement stream

(draft : “Block Announcements”, kind 0x00). The header is abstracted to the height it carries.

$\text{MakeHeaderAnnouncement}(\text{height}) \triangleq [$
 $\quad \text{kind} \quad \mapsto \text{“header”},$
 $\quad \text{height} \mapsto \text{height}$
 $]$